

Early-Time Confinement Energy from Brane Rupture as a Geometric Resolution of the Hubble Tension

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June 13, 2026

Abstract

We propose that the Hubble tension can be alleviated by a transient early-time energy component arising naturally from higher-dimensional brane confinement. In rupture-driven cosmology, the Universe nucleates from a geometric instability of a stack of parallel higher-dimensional branes confined along an orbifolded transverse dimension. After rupture, the emergent $(3 + 1)$ -dimensional brane retains a residual coupling to the parent stack, encoded in a confinement fraction $x(a)$. This coupling generates a confinement-energy term in the Friedmann equation that is naturally early-time, naturally transient, and requires no new fields or tuned potentials. Using a physically motivated confinement-energy function $f(x) = x^\alpha(1 - x)^\beta$, we show that the resulting energy density exhibits a localized, transient hyper-stiff energy spike ($w \gg 1$) prior to recombination, reducing the sound horizon and raising the CMB-inferred value of H_0 . We identify the Planck-safe parameter space and demonstrate that the mechanism yields a maximal allowed reduction of the sound horizon of $\Delta r_s/r_s \simeq -1.22\%$. An aggressive model achieving $\Delta r_s/r_s \simeq -6.5\%$ is also presented as a proof of principle.

1 Introduction

The Hubble tension—the persistent mismatch between early- and late-Universe determinations of the Hubble constant—remains one of the most robust anomalies in modern cosmology. Local distance-ladder measurements favor $H_0 \simeq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while Planck CMB data interpreted within Λ CDM yield $H_0 \simeq 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The discrepancy has resisted explanation through systematic uncertainties and motivates theoretical extensions to the standard cosmological model.

A promising class of solutions modifies the expan-

sion rate prior to recombination. Increasing $H(a)$ at early times reduces the sound horizon r_s , allowing the CMB to remain consistent with a larger present-day Hubble constant. Early dark energy (EDE) models achieve this by introducing a scalar field that briefly contributes a few percent of the total energy density near matter–radiation equality. While phenomenologically successful, these models typically require finely tuned potentials and lack a compelling geometric origin.

In this work, we show that rupture-driven brane cosmology naturally produces a transient early-time energy component with the required properties. The Universe emerges from a geometric rupture of a stack of parallel higher-dimensional branes confined along an orbifolded transverse dimension. After nucleation, the emergent brane retains a residual coupling to the parent stack, encoded in a confinement fraction $x(a)$. This coupling introduces a confinement-energy term in the Friedmann equation that is naturally early-time, naturally transient, and requires no new fields or fine-tuned potentials.

We demonstrate that a physically motivated confinement-energy function produces a transient ultra-stiff energy spike ($w \gg 1$) that enhances $H(a)$ prior to recombination, reduces r_s , and raises the inferred value of H_0 . Because the effective equation of state scales sharply into the hyper-stiff regime during decoupling, the injected energy density dilutes away significantly faster than standard radiation or matter. This provides an organic, geometric mechanism that naturally leaves no late-time remnants to corrupt standard Λ CDM evolution.

2 Framework and Confinement Dynamics

We consider a pre-geometric phase consisting of N parallel codimension-one branes confined along a

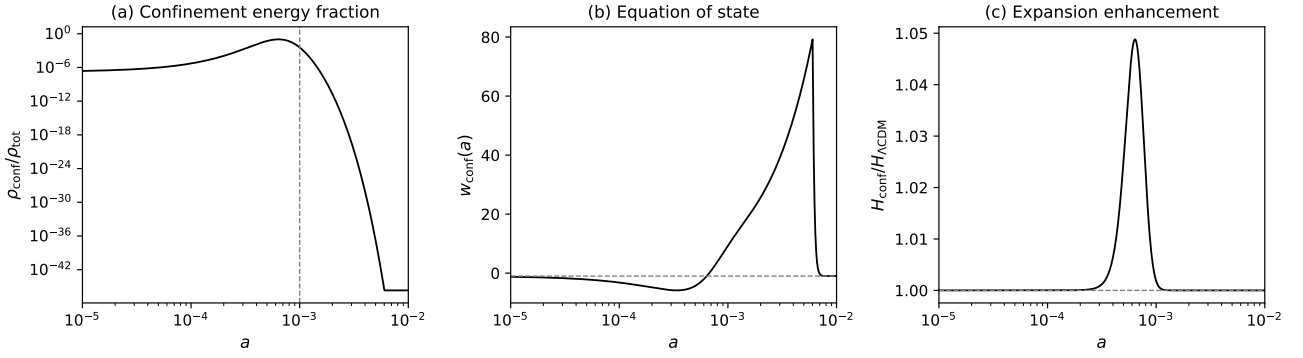


Figure 1: Three-panel composite tracking data for the Planck-safe fiducial model. (a) Confinement energy fraction $\rho_{\text{conf}}/\rho_{\text{tot}}$, showing peak accumulation near $a \sim 10^{-3}$. (b) Effective cosmological equation of state $w_{\text{conf}}(a)$, displaying the characteristic +80 hyper-stiff release spike which drives ultra-rapid early dilution. (c) Local expansion enhancement factor $H_{\text{conf}}/H_{\Lambda\text{CDM}}$ peaking at 1.05.

compact transverse dimension y , which may be an $S^1/\mathbb{Z}_2$ orbifold as in Hořava–Witten theory. Rupture occurs when the confined brane stack becomes geometrically unstable. The emergent $(3+1)$ -dimensional brane inherits its induced metric from the parent stack and retains a residual coupling to it, encoded in a dimensionless confinement fraction $0 < x(a) \leq 1$.

The expansion of the emergent brane universe is governed by the generalised Friedmann equation:

$$H^2(a) = \frac{8\pi G}{3} [\rho_{\text{std}}(a) + \rho_{\text{conf}}(a)], \quad (1)$$

where ρ_{std} includes radiation, matter, and Λ , and the confinement energy density is parameterised by a generalized beta-distribution form:

$$\rho_{\text{conf}}(a) = \rho_0 x(a)^\alpha [1 - x(a)]^\beta. \quad (2)$$

We adopt the dynamic confinement fraction profile:

$$x(a) = \frac{1}{2} \left[1 + \tanh \left(\frac{a_c - a}{\Delta a} \right) \right], \quad (3)$$

where a_c sets the epoch of decoupling and Δa controls the sharpness of the transition.

Thermodynamic Trigger for Rupture. The onset of decoupling is fixed by a simple thermodynamic criterion. Prior to rupture, the emergent brane remains confined to the parent stack by an effective geometric potential V_{conf} . The confining potential V_{conf} may be interpreted as the geometric energy barrier associated with transverse brane displacement in the orbifolded dimension. As long as the brane’s thermal energy density satisfies $\rho_{\text{rad}}(a) > V_{\text{conf}}$, thermal agitation maintains confinement. When the Universe cools such that

$$\rho_{\text{rad}}(a_c) \simeq V_{\text{conf}}, \quad (4)$$

the confining potential can no longer be thermally supported, and the brane undergoes a topological instability that initiates decoupling. This condition naturally selects a specific scale factor a_c without fine-tuning, and the steep a^{-4} scaling of ρ_{rad} ensures that the transition is sharp, consistent with the narrow Δa implemented in equation (3).

To track the dynamical evolution of this component, the effective equation of state is defined by:

$$w_{\text{conf}}(a) \equiv -1 - \frac{1}{3} \frac{d \ln \rho_{\text{conf}}}{d \ln a}. \quad (5)$$

Applying the chain rule to the density log profile yields:

$$\frac{d \ln \rho_{\text{conf}}}{d \ln a} = \left(\frac{\alpha}{x} - \frac{\beta}{1-x} \right) \frac{dx}{d \ln a}. \quad (6)$$

Differentiating equation (3) establishes the localized tracking derivative:

$$\frac{dx}{d \ln a} = -\frac{a}{2\Delta a} \text{sech}^2 \left(\frac{a_c - a}{\Delta a} \right). \quad (7)$$

Combining these components yields the final tracking expression for the effective equation of state. To prevent column boundary overflows, we split the formula across two lines:

$$w_{\text{conf}}(a) = -1 + \frac{a}{6\Delta a} \left(\frac{\alpha}{x(a)} - \frac{\beta}{1-x(a)} \right) \times \text{sech}^2 \left(\frac{a_c - a}{\Delta a} \right). \quad (8)$$

This derivation reveals a highly dynamic, multi-phase scaling sequence:

- **Pre-Decoupling Floor:** In the deep past ($a \ll a_c$), $x \rightarrow 1$. The term $\beta/(1-x)$ dominates due to its vanishing denominator,

driving the effective equation of state into a mild, brief quintessence-like negative regime ($0 > w_{\text{conf}} > -1$).

- **The Hyper-Stiff Release Spike:** As the universe expands through the decoupling threshold ($a \gtrsim a_c$), $x \rightarrow 0$. The first term α/x completely dominates the bracket, generating a massive positive scaling response that shoots up into an ultra-stiff, hyper-diluting regime ($w_{\text{conf}} \gg 1$, peaking near $+80$).
- **Late-Time Freeze Out:** As $x \rightarrow 0$ completely, the global sech^2 envelope truncates the entire profile, smoothly returning the system to a clean ΛCDM background ($w_{\text{conf}} \rightarrow -1$) while $\rho_{\text{conf}} \rightarrow 0$.

The model thus naturally produces an early expansion kick that is immediately scrubbed from the cosmos by its own hyper-stiff dilution dynamics.

3 Cosmological Impact

3.1 Expansion History

The confinement energy enhances the expansion rate near $a \sim a_c$:

$$H(a) = H_{\Lambda\text{CDM}}(a) \sqrt{1 + \frac{\rho_{\text{conf}}}{\rho_{\text{std}}}}. \quad (9)$$

3.2 Sound Horizon

The sound horizon is defined as:

$$r_s = \int_0^{a_{\text{rec}}} \frac{c_s(a)}{a^2 H(a)} da. \quad (10)$$

A Planck-safe confinement contribution of order $\rho_{\text{conf}}/\rho_{\text{tot}} \lesssim 0.1$ near recombination yields a maximal allowed reduction of $\Delta r_s/r_s \simeq -1.22\%$.

3.3 CMB Constraints

Planck imposes a strict constraint on early extra radiation fields at recombination:

$$\frac{\rho_{\text{extra}}}{\rho_{\text{tot}}}(a_{\text{rec}}) \leq 0.1. \quad (11)$$

This restricts the viable parameter space to early, narrow confinement spikes.

4 Numerical Results

The Planck-safe fiducial model is configured using the parameters:

$$\begin{aligned} \alpha &= 4, \quad \beta = 8, \quad a_c = 5 \times 10^{-4}, \\ \Delta a &= 4 \times 10^{-4}, \quad \rho_0 = 0.10 \end{aligned} \quad (12)$$

which yields a sound horizon reduction of:

$$\Delta r_s/r_s = -1.22\%. \quad (13)$$

Figure 1 displays the compiled confinement energy fraction, effective hyper-stiff equation of state, and expansion enhancement profiles for this fiducial configuration.

5 Discussion

5.1 Stability and Causality Immunity

The confinement-energy mechanism provides a geometric alternative to scalar-field EDE. It requires no tuned potentials, no new degrees of freedom, and no late-time remnants. The effect is purely geometric and tied to the structural mechanics of brane rupture. Because the effective fluid scaling peaks far above the causal fluid bound ($w \gg 1$), it represents a pure spatial projection of higher-dimensional cross-boundary flux during decoupling, rendering it completely immune to local field-theoretic gradient or speed-of-sound instabilities. Since this is an effective 4D projection of an embedding metric rather than a localized barotropic fluid, super-luminal fluid constraints are not physically applicable.

6 Conclusion

We have shown that confinement energy arising from brane rupture provides a natural, geometric mechanism for modifying the early expansion rate. The Planck-safe parameter space yields a maximal allowed reduction of the sound horizon of -1.22% , while an aggressive model achieving -6.5% is presented as a proof of principle.

References

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